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EXERCISE 2. SQL AGGREGATES & OPERATORS

(Q1) SELECT DISTINCT department
FROM students;

department
IT
HR
Finance

(Q2) SELECT department,
AVG (age) AS avg-age
FROM students
GROUP BY department;

department	avg-age
IT	20.5
HR	22
Finance	23

(Q3) SELECT department,
COUNT (student-id) AS student-count
FROM students
GROUP BY department
HAVING COUNT (student-id) > 1;

department	student-count
IT	2
HR	2

Q4) SELECT student_id,
name,
age,
department

FROM students

WHERE age BETWEEN 21 AND 23;

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

Q5) SELECT student_id,
name,
age,
department

FROM students

WHERE department IN ('IT', 'HR') AND age > 21;

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

The courses Table

```
(Q6) SELECT department,
      sum(credits) as total_credits
FROM courses
GROUP BY department
HAVING sum(credits) > 5;
```

department	total_credits
IT	11

```
(Q7) SELECT course_id,
      course_name,
      department,
      credits
```

FROM courses

```
WHERE credits <= 4;
```

Course-id	Course-name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	2

```
(Q8) SELECT course_id,
      course_name,
      Credits
```

FROM courses

```
ORDER BY Credits DESC
```

```
Limit 3;
```

Course-id	Course-name	Credits
103	Data Science	4
102	Python	4
101	SQL Basics	3

The enrollments Table

```
(09) SELECT MAX(grade) AS max-grade,  
            MIN(grade) AS min-grade,  
            AVG(grade) AS avg-grade  
FROM enrollments;
```

max-grade	min-grade	avg-grade
90	78	84.6

```
(10) SELECT course-id,  
            COUNT(enrollment-id) AS enrollment-count  
FROM enrollments  
GROUP BY enrollment-id;
```

course-id	enrollment-count
101	1
102	2
103	3
104	4
105	5

The salaries Table

```
(11) SELECT department,  
            SUM(salary) AS total-salary,  
            SUM(bonus) AS total-bonus  
FROM salaries
```

GROUP BY department;

department	total-salary	total-bonus
IT	122,000	10,500
HR	109,000	7,600
Finance	70,000	6,000

(12) SELECT department,
 sum(salary) AS avg_salary
 FROM ~~department~~ salaries
 GROUP BY department
 HAVING avg_salary > 55,000;

department	avg_salary
IT	61000
HR	82000

(13) SELECT employee_id,
 name,
 salary,
 bonus, salary + bonus AS total_compensation
 FROM salaries
 WHERE salary + bonus > 60000;

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

The Projects Table

(14) SELECT department,
 sum(budget) AS total_budget
 Avg(budget) AS ^{avg}total_budget
 FROM projects
 GROUP BY department
 HAVING avg(budget) > 70000;

department	total-budget	avg-budget
IT	870000	130000
Finance	80000	80000

(10) SELECT project-id,
project-name,
department,
budget

FROM projects

WHERE budget BETWEEN 50000 AND 120000

AND department != 'marketing';

project-id	project-name	department	budget
2	Payroll System Finance	Finance	80000
5	HR Portal HR	HR	50000